

Partial random circulant kernels are not orthant symmetric

A probability-one Pluecker obstruction at $M = N/2$

Autonomous mathematical discovery run `fa_banach_001`

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Status. Candidate full negative answer, likely valid pending expert review.

1 Source question

Keiper studies partial random circulant measurement matrices generated by a centered sub-Gaussian vector. In Section 4 of [1], PDF pages 13–14, she asks whether the kernel of such a matrix is orthant symmetric. In the paper’s formulation, if the rows of a random matrix B span the kernel, one would require

$$\mathbb{P}(BS \in \Omega) = \mathbb{P}(B \in \Omega)$$

for every diagonal sign matrix S and measurable set Ω . In particular, this condition forces the distribution of the random kernel subspace itself to be invariant under $K \mapsto KS$.

Note, that Φ_Θ is in general position if for every subset $B \subset [N]$ of cardinality $|B| = M$ the eigenvalues of the matrix $(\Phi_\Theta^B)^* \Phi_\Theta^B$ are positive, i.e., if $\lambda((\Phi_\Theta^B)^* \Phi_\Theta^B) > 0$ or $s_N(\Phi_\Theta^B) \geq 0$, respectively, where $s_N(A)$ denotes the smallest singular value of a matrix $A \in \mathbb{R}^{N,N}$. Note that there exists a lot of work in the literature calculating the probability of $\lambda(\Phi^* \Phi) > 0$ ([9, 14]), but an analogous calculation for $\lambda((\Phi^B)^* \Phi^B)$ is much more difficult. The main reason is that the eigenvalues of Φ itself can be computed very easily, which is not the case for a submatrix. In particular it has been shown in [14], for Rademacher sequences $(b_0, \dots, b_{N-1})$ (among others) that for all $\varepsilon > 0$ and large N

$$\mathbb{P}(s_N(\Phi(b)) \geq \varepsilon N^{-1}) \geq 1 - C\varepsilon, \quad (13)$$

where $C > 0$ is a constant only depending on b . Since Φ_Θ^B seems to be more unstructured than Φ itself one might hope that the probability of $\lambda((\Phi_\Theta^B)^* \Phi_\Theta^B) > 0$ is even higher.

However, the proof of Theorem III.2 in [5], is also based upon the fact that the random part Φ of the measurement matrix $A = \mu \mathbb{1} + \Phi$ has an orthant symmetric kernel. This means, if for each diagonal matrix $S \in \mathbb{R}^{N,N}$ with diagonal in $\{-1, 1\}^N$ and every measurable set $\Omega \in \mathbb{R}^{M,N}$, it holds $\mathbb{P}(BS \in \Omega) = \mathbb{P}(B \in \Omega)$, where B is a matrix whose rows span the kernel of Φ .

From our point of view, it seems to be very unlikely that this is true for partial circulant matrices. The reason is the following. Suppose the kernel of Φ_Θ is spanned by $v_1, \dots, v_m$, for some $m \in [N]$, and let $B \in \mathbb{R}^{m,N}$ the matrix with rows $v'_1, \dots, v'_m$, then the rows of BS span the kernel of TS . But TS is in general no circulant matrix.

So for future work it is interesting to check if the kernel of a partial circulant matrix is indeed not or perhaps is orthant symmetric. Independently of the answer to this question it might be of own interest to prove an inequality in the spirit of (13) for partial circulant matrices. And, if the answer to the first question is negative, to find an alternative way to proof the upper bound on the necessary number of measurements in the order of $M \lesssim N/2$.

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The source suspects that the answer is negative and specifically connects the question to the balanced recovery threshold $M \lesssim N/2$. The example below gives a negative answer already at $M = N/2$.

2 Proof intuition

The rows of a 2×4 partial circulant matrix depend on four random parameters, but their two-plane is not a generic point of the Grassmannian. Its six Pluecker coordinates always satisfy an additional quadratic equation $Q = 0$. This equation is intrinsic to the plane because it is homogeneous in the Pluecker coordinates.

A single coordinate sign flip changes three of those six coordinates. The same quadratic then becomes a nonzero polynomial in the four generating variables. With a Gaussian input, a nonzero polynomial vanishes with probability zero. Thus an algebraic event has probability one for the original row space and zero for its sign flip. Orthogonal complementation turns the row-space obstruction into the desired kernel obstruction.

3 Counterexample

For a two-plane $W \subset \mathbb{R}^4$ with ordered basis u, v , write

$$p_{ij}(W) = u_i v_j - u_j v_i \quad (0 \leq i < j \leq 3)$$

for its Pluecker coordinates. A different ordered basis multiplies all six coordinates by the same nonzero scalar. Consequently, the zero set of any homogeneous polynomial in the p_{ij} is a well-defined event on the Grassmannian.

Theorem 1. *Let a, b, c, d be independent standard real Gaussian random variables and let*

$$\Phi = \begin{pmatrix} a & b & c & d \\ d & a & b & c \end{pmatrix},$$

the partial circulant matrix formed from rows 0 and 1 of the circulant matrix generated by (a, b, c, d) . Then Φ has rank two almost surely, and the law of $\ker \Phi$ is not invariant under coordinate sign changes. More strongly, for $S = \text{diag}(-1, 1, 1, 1)$, the laws of $\text{row}(\Phi)$ and $\text{row}(\Phi)S$ are separated by a measurable algebraic event having probabilities one and zero.

Proof. The two rows fail to be independent only on the common zero set of the six 2×2 minors. This set has Gaussian probability zero, since, for example, the minor $a^2 - bd$ is a nonzero polynomial. Hence the row space is a two-plane almost surely.

Its six Pluecker coordinates are

$$\begin{aligned} p_{01} &= a^2 - bd, & p_{02} &= ab - cd, & p_{03} &= ac - d^2, \\ p_{12} &= b^2 - ac, & p_{13} &= bc - ad, & p_{23} &= c^2 - bd. \end{aligned}$$

Define the homogeneous quadratic

$$Q(p) = (p_{01} - p_{23})(p_{03} + p_{12}) - p_{02}^2 + p_{13}^2. \quad (1)$$

For the coordinates above,

$$p_{01} - p_{23} = a^2 - c^2, \quad p_{03} + p_{12} = b^2 - d^2.$$

Moreover,

$$\begin{aligned} p_{02}^2 - p_{13}^2 &= (ab - cd - bc + ad)(ab - cd + bc - ad) \\ &= (a - c)(b + d)(a + c)(b - d) \\ &= (a^2 - c^2)(b^2 - d^2). \end{aligned}$$

Therefore

$$Q(p(\text{row}(\Phi))) = 0 \tag{2}$$

identically. Let $\mathcal{A}$ be the algebraic subset of the Grassmannian defined by $Q = 0$. Equation (2) says $\mathbb{P}\{\text{row}(\Phi) \in \mathcal{A}\} = 1$.

Now right multiplication by $S = \text{diag}(-1, 1, 1, 1)$ changes the sign of p_{01}, p_{02}, p_{03} and leaves p_{12}, p_{13}, p_{23} unchanged. Thus

$$Q(p(\text{row}(\Phi)S)) = Q(-p_{01}, -p_{02}, -p_{03}, p_{12}, p_{13}, p_{23}) \tag{3}$$

is a polynomial in a, b, c, d . It is not the zero polynomial: at $(a, b, c, d) = (1, 0, 1, 0)$ its value is 4. The zero set of a nonzero real polynomial has Lebesgue measure zero, and the Gaussian law has a density. Consequently,

$$\mathbb{P}\{\text{row}(\Phi)S \in \mathcal{A}\} = 0.$$

The row-space law is therefore not sign-invariant.

Finally, S is orthogonal and

$$(\ker \Phi S)^\perp = (\ker \Phi)^\perp S = \text{row}(\Phi)S.$$

If the kernel law were invariant under $K \mapsto KS$, then its orthogonal complement, the row-space law, would also be invariant. The preceding probability-one/probability-zero separation contradicts this. Hence the kernel is not orthant symmetric. $\square$

Remark 2 (Distributional scope). *The proof only uses that (a, b, c, d) has an absolutely continuous joint law; independence and Gaussianity are a natural match to the source but are not needed for the algebraic separator. Thus the same conclusion holds for every absolutely continuous generating-vector distribution on $\mathbb{R}^4$.*

4 Independent small-dimensional check

There is also a direct 2×3 witness. For iid standard Gaussians b_0, b_1, b_2 , the kernel of

$$\begin{pmatrix} b_0 & b_1 & b_2 \\ b_2 & b_0 & b_1 \end{pmatrix}$$

is spanned almost surely by

$$v = (b_1^2 - b_0b_2, b_2^2 - b_0b_1, b_0^2 - b_1b_2).$$

The identity

$$v_0 + v_1 + v_2 = \frac{1}{2}((b_0 - b_1)^2 + (b_1 - b_2)^2 + (b_2 - b_0)^2) > 0$$

holds almost surely. Hence the kernel line meets the positive orthant exactly when all three coordinates of v are positive. On the two input orthants in which the b_j have a common sign this

is impossible, because multiplying the three required strict inequalities gives equality. On each of the six mixed sign orthants, two inequalities are automatic and the third is $X^2 > YZ$ for iid half-normal magnitudes. Therefore

$$\mathbb{P}\{\ker \Phi \text{ meets } \mathbb{R}_{>0}^3\} = \frac{3}{4} \mathbb{P}\{X^2 > YZ\} > \frac{3}{4} \cdot \frac{1}{3} = \frac{1}{4}.$$

The strict inequality follows because $\{X > \max(Y, Z)\}$ has probability $1/3$, is contained in $\{X^2 > YZ\}$, and the containment is strict on, for example, the positive probability box

$$1.5 < X < 1.6, \quad 2 < Y < 2.1, \quad 0.5 < Z < 0.6.$$

An orthant-invariant random line with no zero coordinates assigns probability $1/4$ to each of the four projective orthants, so this yields a second contradiction. The main theorem is preferable because it lies exactly at $M = N/2$ and gives a probability-one separator.

5 Verification, limitations, and novelty

The accompanying script symbolically expands (1), verifies its identical vanishing before the sign flip, and verifies that the flipped polynomial takes value 4 at the displayed witness. These checks are not part of the proof.

The theorem gives a full negative answer to the universal orthant-symmetry question for the Gaussian partial-circulant ensemble considered by the source, and it does so in the balanced regime $M = N/2$. It does not classify every dimension and row subset, does not settle a fixed-rank Rademacher formulation, and does not provide the alternative route to the recovery upper bound which the source requests after a negative answer.

A bounded novelty search on 17 August 2026 covered the four run indexes, the local parsed source, the current arXiv record, exact searches for the source question and arXiv id, partial-circulant/orthant-symmetry variants, and the 2×3 cross-product expression. No later answer or this Pluecker support obstruction was found. Novelty confidence is moderate pending specialist review.

Human review recommendation. Check the six minors, the factorization proving $Q = 0$, the induced Pluecker signs under S , and the orthogonal-complement transfer. The proof uses only elementary algebra and the standard zero-set property of nonzero polynomials.

References

- [1] S. Keiper, *Recovery of Binary Sparse Signals from Structured Biased Measurements*, arXiv:2006.14835 (2020).